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LAB 10	
Aim:	**Mini project:** Design the event driven solution for the real-life scenario under consideration
Theory:	the event-driven online shopping system reflects the core principles behind many real-world software applications, particularly those involving user interfaces and interactive services. In this system, events are primarily triggered by user inputs, which simulate real-time decisions a customer might make while shopping online. For instance, selecting the option to view available products acts as an event that initiates the display of product information, while choosing to checkout triggers calculations of the total cost and simulates the end of a transaction. This interaction model mirrors that of web applications where users click buttons or links, and the system responds accordingly. The backbone of the application lies in its object-oriented structure, where each Product is modelled as a class with its unique attributes (like ID, name, and price), and CartItem manages product quantities within a shopping cart. The OnlineShoppingSystem class acts as the central controller that processes user-generated events. This separation of concerns ensures modularity, making the system more manageable and scalable. An important aspect of this project is the use of control structures (loops and switch-case statements) that continuously monitor for user inputs, much like an event listener in GUI or web applications. This makes the system reactive, responding to each specific user command with the appropriate function. Additionally, the use of collections like ArrayList allows dynamic handling of data such as products and cart items, reinforcing real-world applicability where users may add or remove products multiple times. this event-driven architecture is well-suited for transition into more advanced platforms like graphical user interfaces (GUI) using JavaFX or web applications using technologies like JavaScript and React, which heavily rely on event listeners and state management. Thus, this mini project not only serves as a practical introduction to event-driven programming but also prepares the foundation for building more complex, real-world interactive systems. It encourages students to think in terms of real-time responsiveness, modularity, and user-centric design, which are essential in modern software development.

Program:	//Sheikh Wasimuddin 23071044 CSEIOT-73
	import java.util.*;
	
	class Product {
	  private final int productId;
	  private final String productName;
	  private final double price;
	
	  public Product(int productId, String productName, double price) {
	  this.productId = productId;
	  this.productName = productName;
	  this.price = price;
	  }
	
	  public int getProductId() {

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```
        return productId;
    }

    public String getProductName() {
        return productName;
    }

    public double getPrice() {
        return price;
    }

    @Override
    public String toString() {
        return productId + ". " + productName + " - $" + price;
    }
}

class CartItem {
    private final Product product;
    private int quantity;

    public CartItem(Product product) {
        this.product = product;
        this.quantity = 1;
    }

    public Product getProduct() {
        return product;
    }

    public int getQuantity() {
        return quantity;
    }

    public void incrementQuantity() {
        quantity++;
    }

    public void decrementQuantity() {
        if (quantity > 1) {
            quantity--;
        }
    }

    public double getTotalPrice() {
        return product.getPrice() * quantity;
    }

    @Override
    public String toString() {
        return product.getProductName() + " (x" + quantity + ") - $" + getTotalPrice();
    }
}
```

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```
public class OnlineShoppingSystem {

    private final List<Product> products;
    private final List<CartItem> cart;

    public OnlineShoppingSystem() {
        products = new ArrayList<>();
        cart = new ArrayList<>();

        // Adding some sample products
        products.add(new Product(1, "Laptop", 799.99));
        products.add(new Product(2, "Smartphone", 499.99));
        products.add(new Product(3, "Headphones", 89.99));
        products.add(new Product(4, "Smartwatch", 149.99));
    }

    public void viewProducts() {
        System.out.println("\n--- Available Products ---");
        for (Product product : products) {
            System.out.println(product);
        }
    }

    public void addToCart(int productId) {
        Product product = findProductById(productId);
        if (product != null) {
            // Check if product is already in the cart
            for (CartItem item : cart) {
                if (item.getProduct().getProductId() == productId) {
                    item.incrementQuantity();
                    System.out.println(product.getProductName() + " added to cart. Quantity: " +
item.getQuantity());
                    return;
                }
            }
            // Product is not in the cart yet, add it
            cart.add(new CartItem(product));
            System.out.println(product.getProductName() + " added to cart.");
        } else {
            System.out.println("Product not found.");
        }
    }

    public void viewCart() {
        if (cart.isEmpty()) {
            System.out.println("Your cart is empty.");
        } else {
            System.out.println("\n--- Your Cart ---");
            for (CartItem item : cart) {
                System.out.println(item);
            }
        }
    }
```

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```
    public void removeFromCart(int productId) {
        Iterator<CartItem> iterator = cart.iterator();
        while (iterator.hasNext()) {
            CartItem item = iterator.next();
            if (item.getProduct().getProductId() == productId) {
                item.decrementQuantity();
                if (item.getQuantity() == 0) {
                    iterator.remove();
                }
                System.out.println("Removed one " + item.getProduct().getProductName() + " from
cart.");
                return;
            }
        }
        System.out.println("Product not found in the cart.");
    }

    public void checkout() {
        if (cart.isEmpty()) {
            System.out.println("Your cart is empty. Cannot checkout.");
            return;
        }

        double totalAmount = 0;
        System.out.println("\n--- Checkout ---");
        for (CartItem item : cart) {
            totalAmount += item.getTotalPrice();
        }
        System.out.println("Total amount: $" + totalAmount);
        System.out.println("Thank you for shopping with us!");
    }

    private Product findProductById(int productId) {
        for (Product product : products) {
            if (product.getProductId() == productId) {
                return product;
            }
        }
        return null;
    }

    public static void main(String[] args) {
        OnlineShoppingSystem shoppingSystem = new OnlineShoppingSystem();
        Scanner scanner = new Scanner(System.in);
        int choice;

        do {
            System.out.println("\n===== Online Shopping Menu =====");
            System.out.println("1. View Products");
            System.out.println("2. Add Product to Cart");
            System.out.println("3. View Cart");
            System.out.println("4. Remove Product from Cart");
            System.out.println("5. Checkout");
```

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```
            System.out.println("6. Exit");
            System.out.print("Enter your choice: ");
            choice = scanner.nextInt();

            switch (choice) {
                case 1 -> shoppingSystem.viewProducts();
                case 2 -> {
                    System.out.print("Enter product ID to add to cart: ");
                    int productId = scanner.nextInt();
                    shoppingSystem.addToCart(productId);
                }
                case 3 -> shoppingSystem.viewCart();
                case 4 -> {
                    System.out.print("Enter product ID to remove from cart: ");
                    int productId = scanner.nextInt();
                    shoppingSystem.removeFromCart(productId);
                }
                case 5 -> shoppingSystem.checkout();
                case 6 -> System.out.println("Thank you for using the Online Shopping System!");
                default -> System.out.println("Invalid choice. Try again.");
            }

        } while (choice != 6);

        scanner.close();
    }
}
```

Output:

```
===== Online Shopping Menu =====
1. View Products
2. Add Product to Cart
3. View Cart
4. Remove Product from Cart
5. Checkout
6. Exit
Enter your choice: 1

--- Available Products ---
1. Laptop - $799.99
2. Smartphone - $499.99
3. Headphones - $89.99
4. Smartwatch - $149.99

===== Online Shopping Menu =====
1. View Products
2. Add Product to Cart
3. View Cart
4. Remove Product from Cart
5. Checkout
6. Exit
Enter your choice: 2
Enter product ID to add to cart: 1
Laptop added to cart.
```

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```
===== Online Shopping Menu =====
1. View Products
2. Add Product to Cart
3. View Cart
4. Remove Product from Cart
5. Checkout
6. Exit
Enter your choice: 2
Enter product ID to add to cart: 1
Laptop added to cart. Quantity: 2

===== Online Shopping Menu =====
1. View Products
2. Add Product to Cart
3. View Cart
4. Remove Product from Cart
5. Checkout
6. Exit
Enter your choice: 2
Enter product ID to add to cart: 2
Smartphone added to cart.
```

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```
===== Online Shopping Menu =====
1. View Products
2. Add Product to Cart
3. View Cart
4. Remove Product from Cart
5. Checkout
6. Exit
Enter your choice: 2
Enter product ID to add to cart: 2
Smartphone added to cart.

===== Online Shopping Menu =====
1. View Products
2. Add Product to Cart
3. View Cart
4. Remove Product from Cart
5. Checkout
6. Exit
Enter your choice: 3

--- Your Cart ---
Laptop (x2) - $1599.98
Smartphone (x1) - $499.99
```

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	``` ===== Online Shopping Menu ===== 1. View Products 2. Add Product to Cart 3. View Cart 4. Remove Product from Cart 5. Checkout 6. Exit Enter your choice: 4 Enter product ID to remove from cart: 1 Removed one Laptop from cart. ===== Online Shopping Menu ===== 1. View Products 2. Add Product to Cart 3. View Cart 4. Remove Product from Cart 5. Checkout 6. Exit Enter your choice: 5 --- Checkout --- Total amount: $1299.98 Thank you for shopping with us! ```
Conclusion	Mini Project on event driven solution for the real-life scenario under consideration in java executed successfully.
Name Of Faculty:	Prof. Akhil Jajulwar